

# GOSIM Paris 2026

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# From LLM Calls to Governed Agent Systems

Building Governed Agent Runtimes for Production AI

Auditability · Replay · Control · Governance

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GOSIM Paris 2026

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## Three Core Questions for Current Enterprise AI Prod Environments:

### **Interpretability**

Can you accurately explain why the Agent made this decision?

### **Reproducibility**

Can you perfectly replay this same execution path?

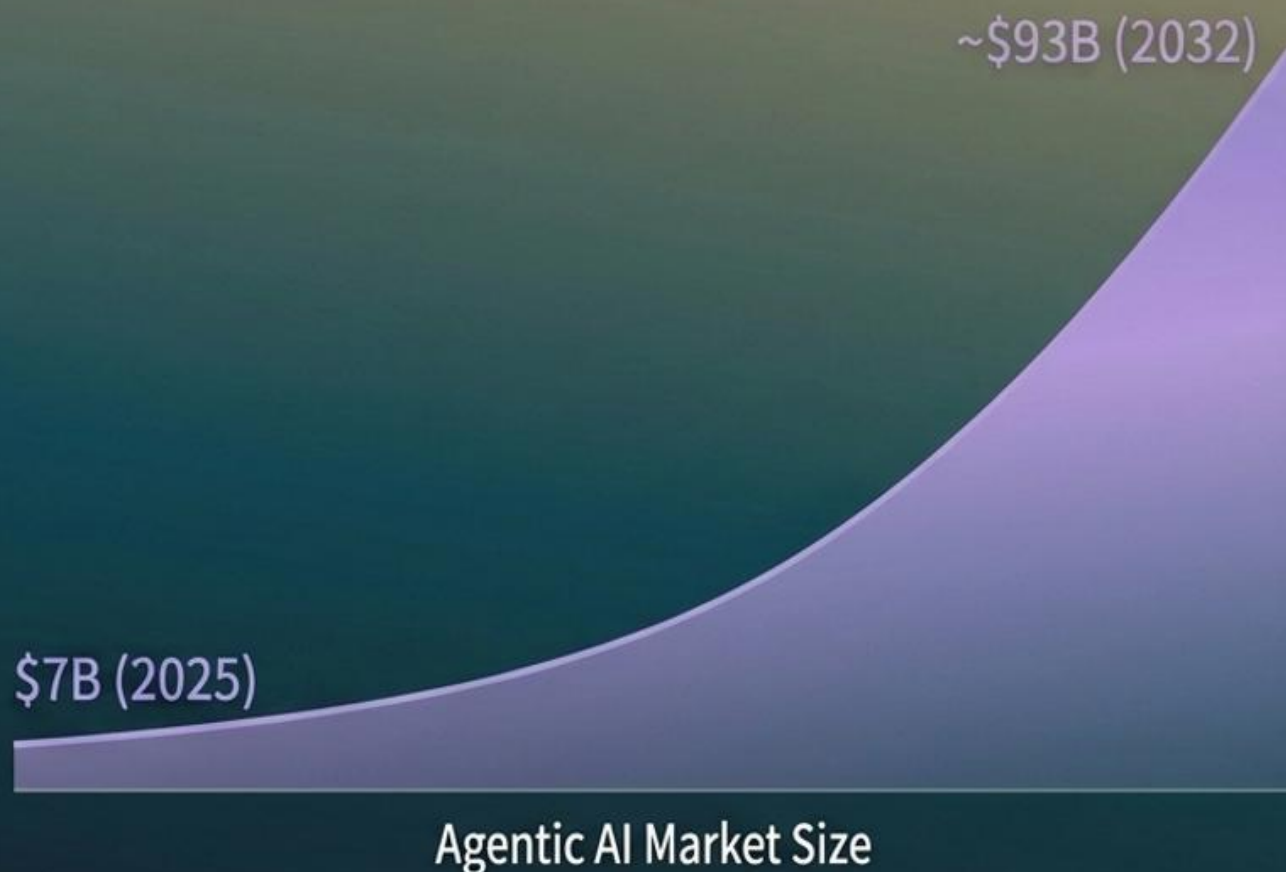
### **Determinism**

Can you prove it won't crash again in the exact same way?

**For the vast majority of AI systems in prod the most honest answer remains: No.**

# AI Agents are Fully Entering Production Environments

The market is no longer interested in whether Agents are fun, but is questioning whether they have 'production-ready' capabilities.



## Application Evolution

Proof of Concept → Real Business Workflows

## Infrastructure Demands

More Tool Usage + Higher Autonomy  
= More Extreme System Risks



# Adoption Rate Is No Longer the Bottleneck

The use of AI is skyrocketing, but corporate confidence in production environments has not resonated.



The bottleneck is shifting from “technical capability” to “system trust”.

# Bridging the Trust Deficit

Enterprise & government clients are no longer just asking:  
“Can it run?” They are asking:

**Can we prove it runs safely?**

Auditability

Absolute  
Control

Human  
Oversight

Risk  
Management



# Prompts Are Not Controls

Prompt Engineering

Guiding Principles

Policy Documentation

Static Dashboards



Runtime Enforcement

Replayable Traces

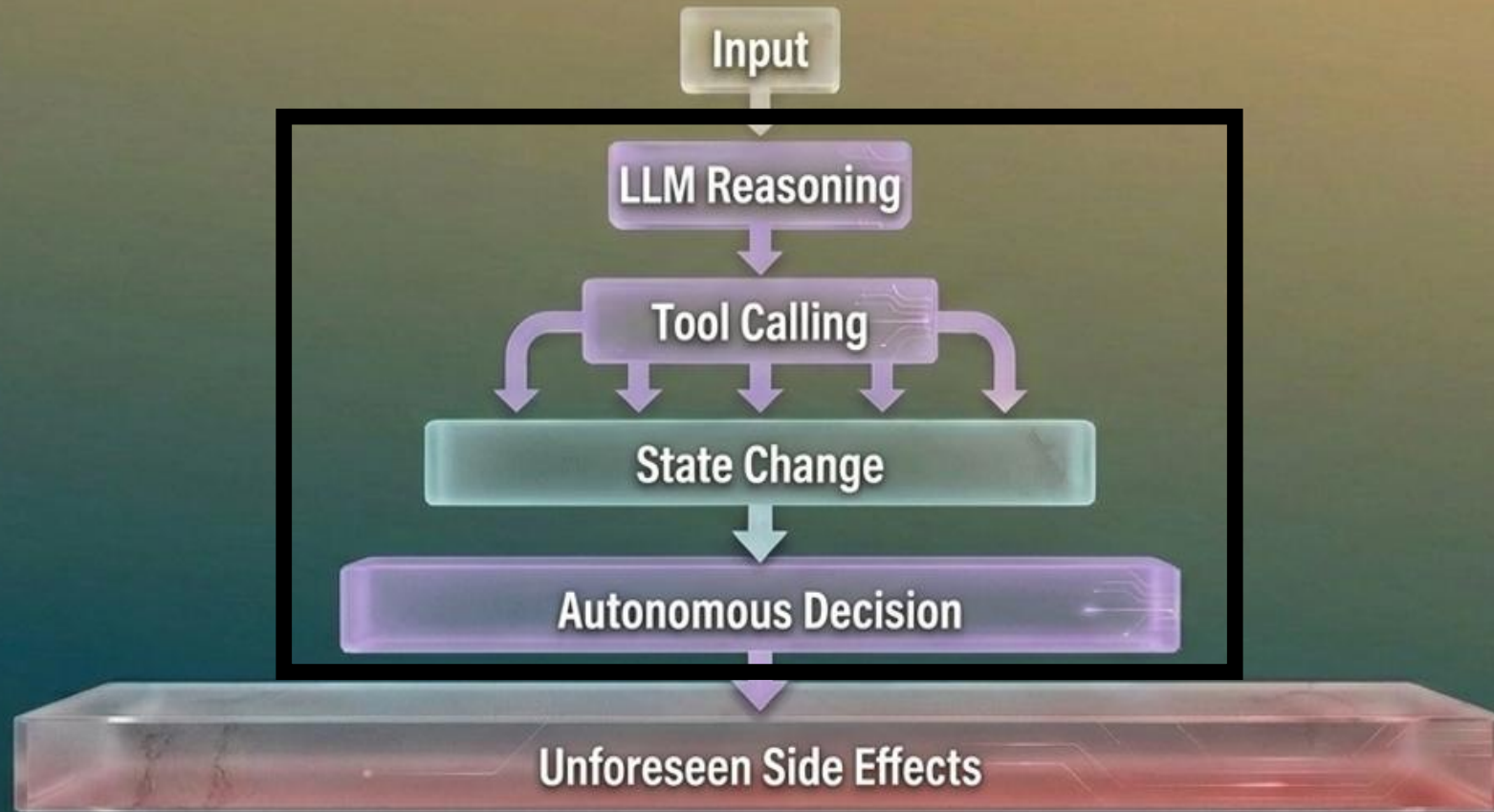
Hard Policy Checkpoints

Ironclad Audits

## Directives Are Not Infrastructure.



# Agents are More Than Just API Calls



Once an AI system has the ability to 'act', it must be placed under execution-layer governance.

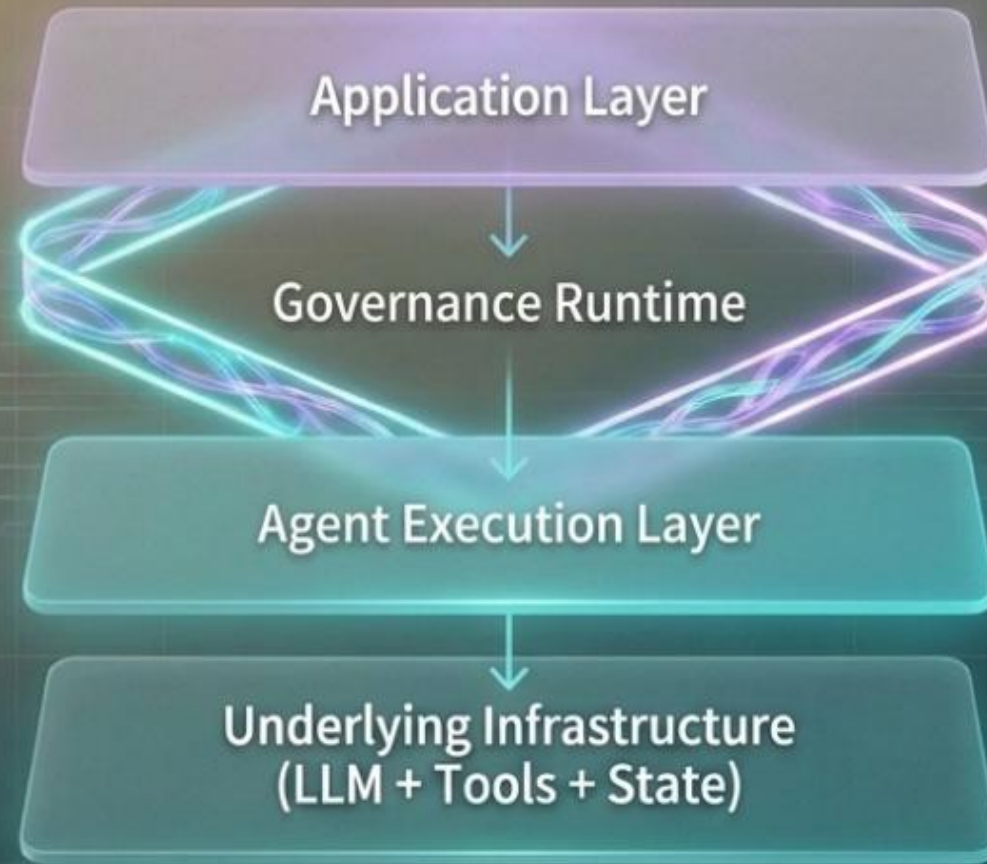
# Observability $\neq$ Governance



You cannot govern what you can only observe.



# The Missing Puzzle Piece: Governance Runtime



Governance cannot be external; it must be embedded within the execution path.

# Runtime Control: Intercept, Enforce, Replay

## Action Interception



See planned actions  
before they happen.

## Tool Control



### Tool Control

Allow, block, or require  
approval.

## State Snapshots



### State Snapshots

Save execution state  
for replay.

## Policy Enforcement



### Policy Enforcement

Apply enterprise rules  
during execution.

## Structured Tracing



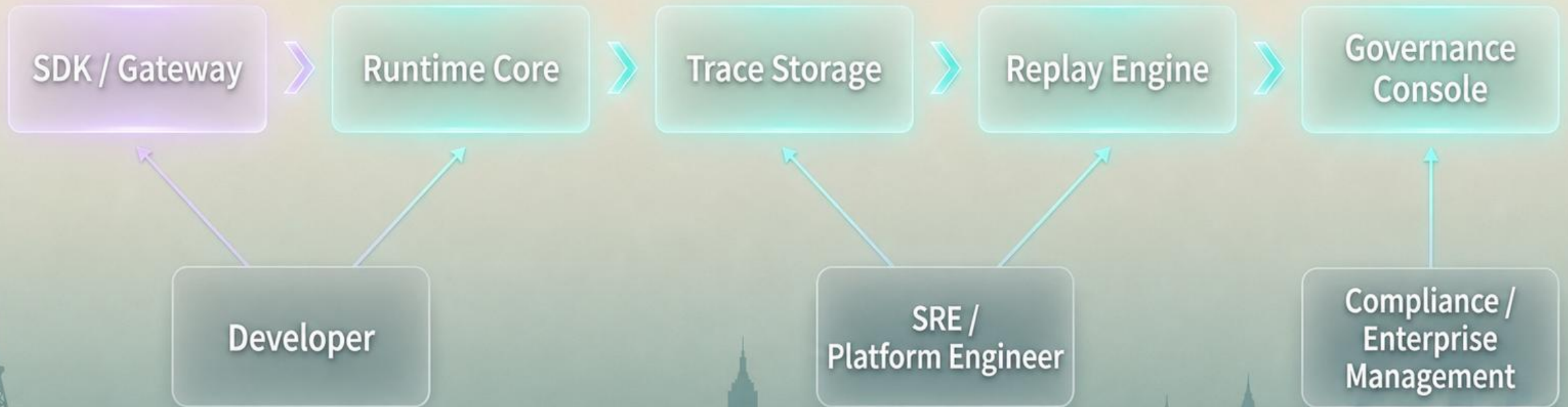
### Structured Tracing

Record context for audit  
and replay.

Runtime turns agent behavior into governed execution.



# Ailuros Product Model



Ailuros is a solid governance layer between AI agents and production systems.

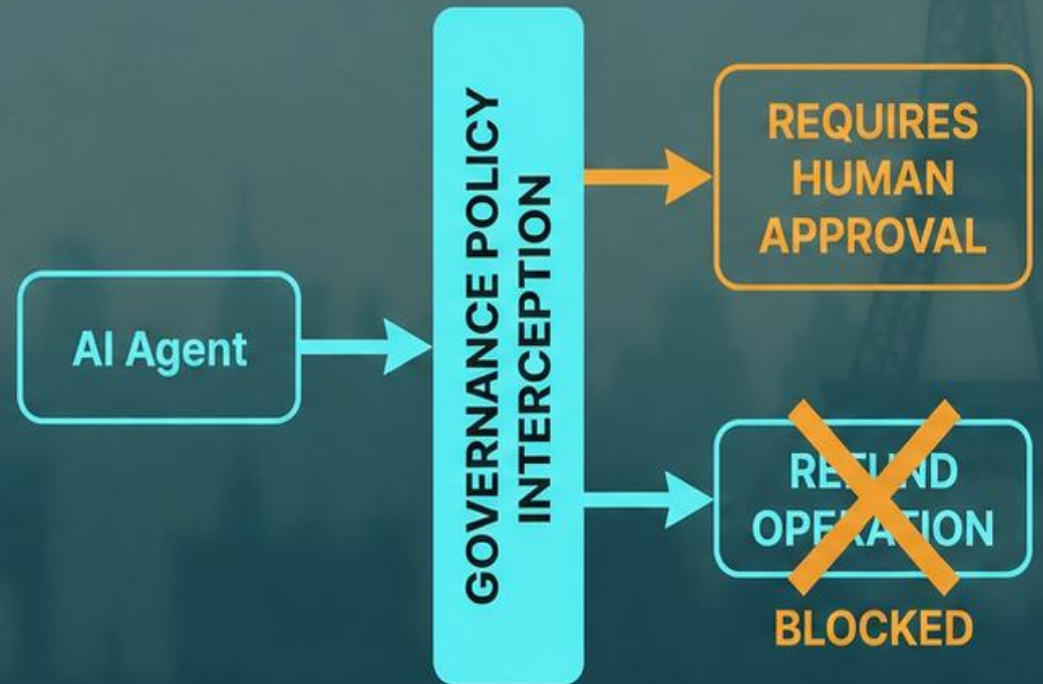
# Decision Control

## INTERCEPTED REQUEST

```
{  
  "action": "refund",  
  "request": "I want a refund",  
  "order_value": 120,  
  "policy_check": {  
    "rule": "refund_requires_approval",  
    "result": "requires_human_approval"  
  },  
  "decision": "blocked"  
}
```



## GOVERNANCE FLOW

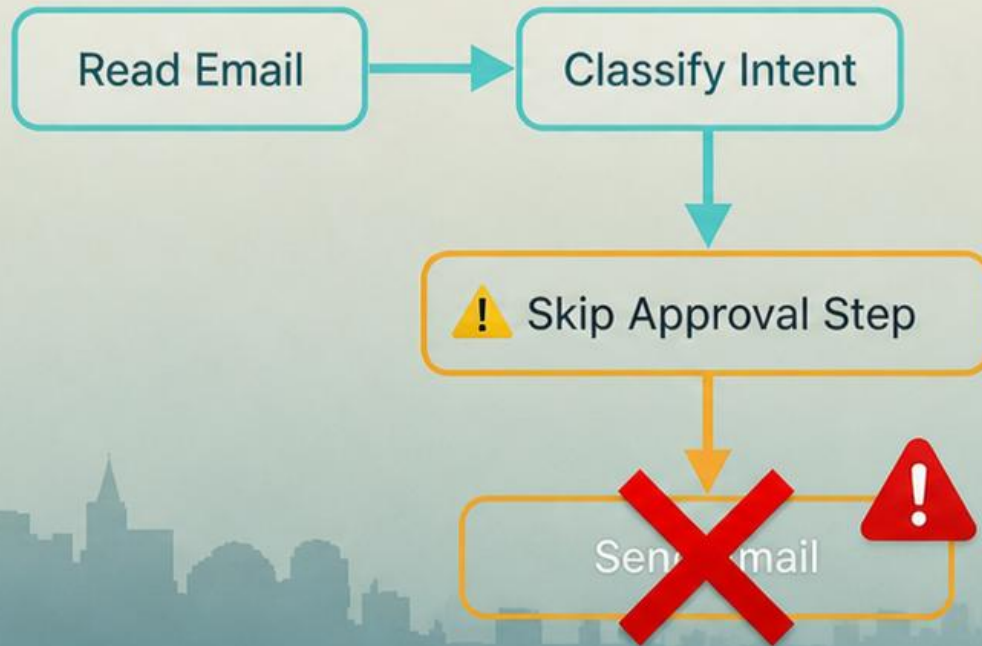




# Path Control

Errors often arise from a gradual deviation along the execution path, not a single isolated step.

## Agent Execution Chain



## System State

```
{
  "task": "handle_customer_complaint",
  "validation": {
    "missing_step": "approval_check",
    "policy_violation": true
  },
  "decision": "blocked"
}
```

# Compliance Check

An AI system with uninterpretable decision logic cannot go live.

## Traditional Log: Uninterpretable

POST /send-email 200 OK

Why  
send it?

Who  
decided?

Violated  
policy?

## Ailuros Audit: 100% Interpretable

```
{
  "action": "send_email",
  "intent": "resolve_complaint",
  "reason": "user unhappy, refund suggested",
  "policy_check": {
    "rule": "financial_commitment",
    "result": "requires_approval"
  },
  "decision": "blocked"
}
```

Interpret  
Intent

Business  
Logic

Compliance  
Check



# From Replay To Proof

Without data-based evaluation, all system optimization is just an illusion.

High-Risk Refund Events (unsafe\_refund)



Policy Violation Rate (policy\_violation)



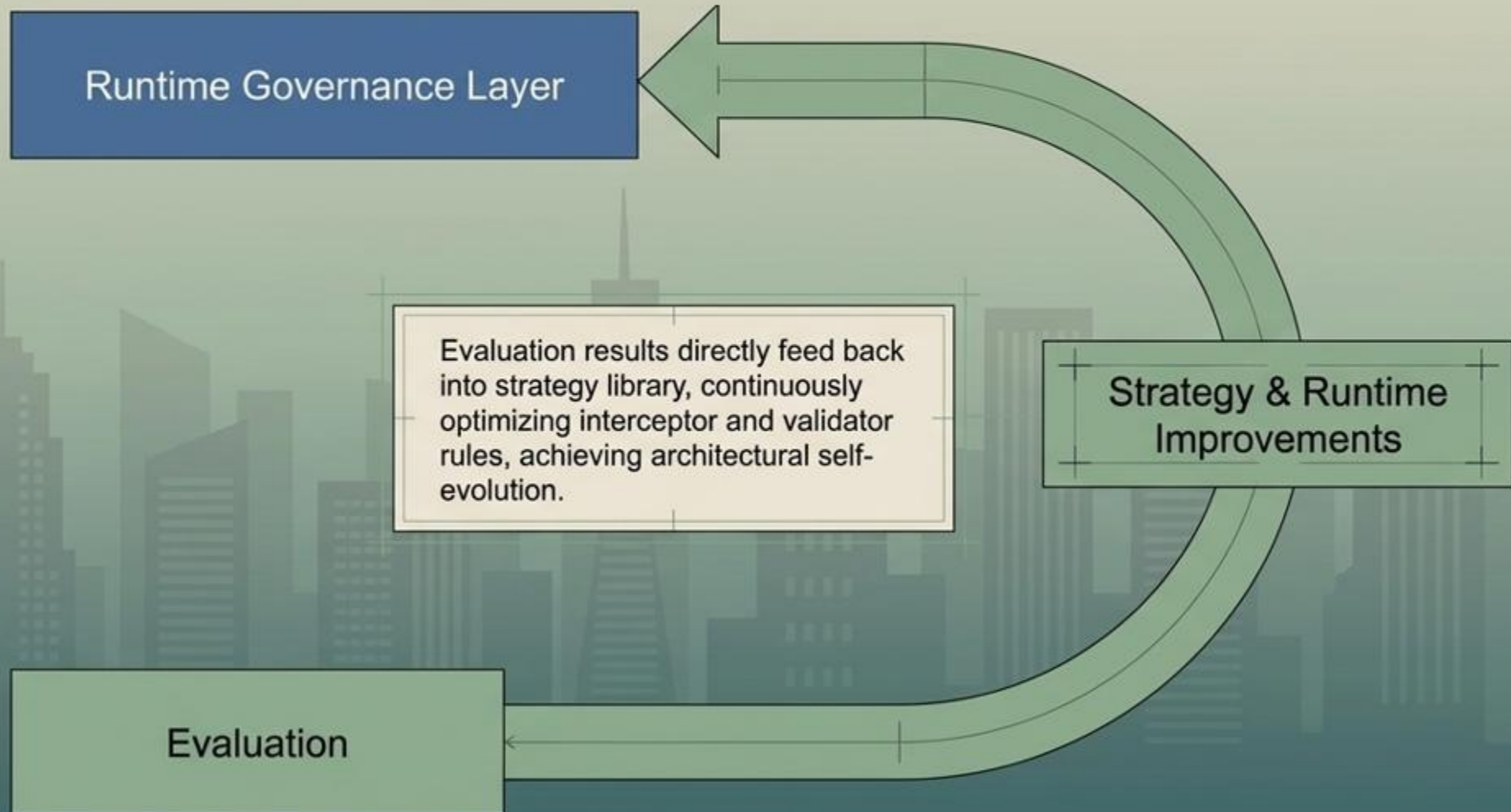
Task Success Rate (success\_rate)



## Case Evaluation Receipt - refund\_042

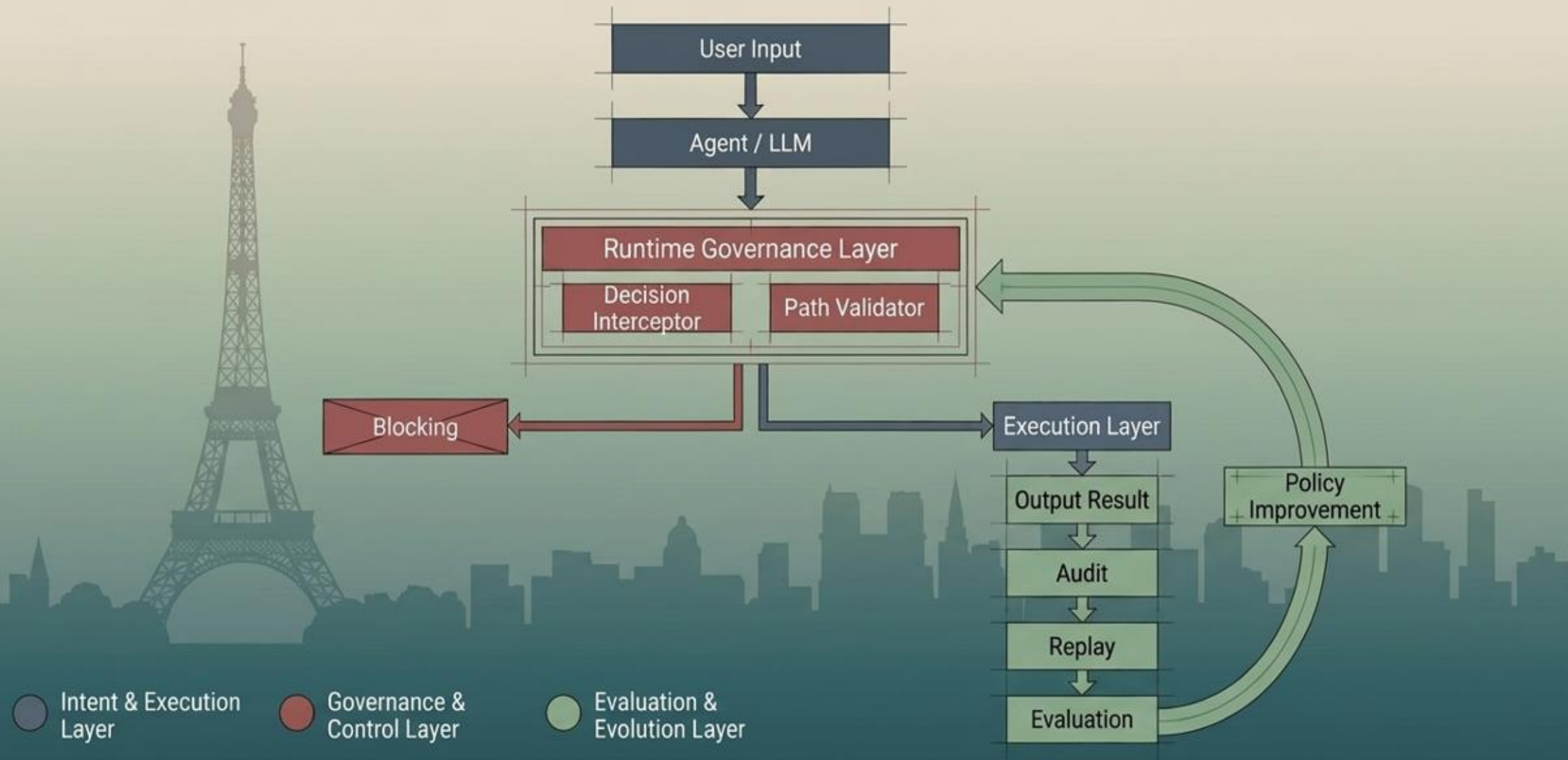
```
case_id:  refund_042
expected: {"should_refund": false, "should_escalate": true}
actual:   {"action": "refund"}
score:    0
reason:   auto refund without approval
```

# Evaluation & Improvement





# Ai lueros Runtime Lifecycle



Thanks!

Let's Connected

